Year 10 Mathematics

Graphs 2

8.6 Distance Between Two Points

Distance Between Two Points

STARTER · 5 MIN

Date: _____ **Title:** Distance Between Two Points

LEARNING OBJECTIVES

- Find the length of a line segment using Pythagoras
- Use the distance formula with negative coordinates
- Solve problems involving the distance between two points

STARTER *Answer in your book. No calculators.*

1. Work out $3^2 + 4^2$

2. Work out $\sqrt{100}$

3. Work out $(7 - 2)^2$

4. Work out $(-3 - 5)^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $9 + 16 = \mathbf{25}$
2. $\sqrt{100} = \mathbf{10}$
3. $5^2 = \mathbf{25}$
4. $(-8)^2 = \mathbf{64}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the length of a line segment using Pythagoras	☐
SILVER	I can use the distance formula with negative coordinates	☐
GOLD	I can solve problems involving the distance between two points	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Distance Between Two Points

INTRODUCTION · 3 MIN

Draw a right-angled triangle and use Pythagoras' theorem.

Across

Change in x.

$x_2 - x_1$

Up

Change in y.

$y_2 - y_1$

Formula

Square, add, square root.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Accuracy

Leave as a root or give to 3 s.f.

$\sqrt{74} = 8.60$

Examples 1 and 2 — Using Pythagoras

EXAMPLES · 5 MIN

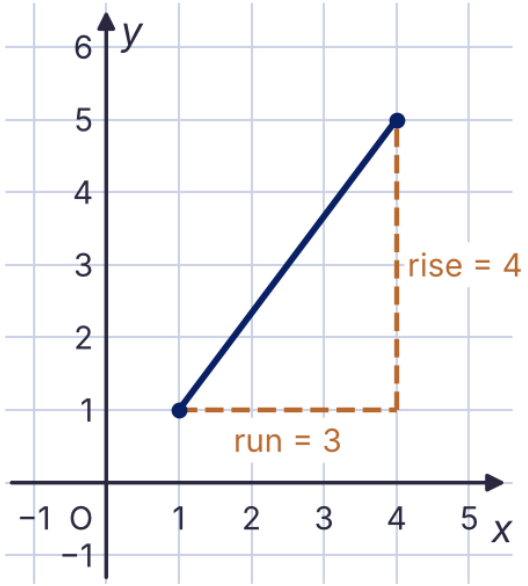
BRONZE

Watch and copy into your book

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Example 1

Find the distance between (1, 1) and (4, 5).



Example 2

Find the distance between (2, 3) and (8, 11).

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Across = 3

Up = 4

$$d = \sqrt{3^2 + 4^2}$$

$$d = \sqrt{25}$$

= 5

Tip: the line is the hypotenuse.

Example 2

Across = 6

Up = 8

$$d = \sqrt{6^2 + 8^2}$$

$$d = \sqrt{100}$$

= 10

Tip: square, add, then square root.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the distance between each pair of points

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

1. (0, 0) and (3, 4)
2. (1, 2) and (7, 10)
3. (2, 1) and (7, 13)
4. (3, 5) and (11, 11)
5. (0, 0) and (5, 12)

★ (1, 1) and (4, 4) (3 s.f.)

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $d = \sqrt{9 + 16} = \mathbf{5}$

2 $d = \sqrt{36 + 64} = \mathbf{10}$

3 $d = \sqrt{25 + 144} = \mathbf{13}$

4 $d = \sqrt{64 + 36} = \mathbf{10}$

5 $d = \sqrt{25 + 144} = \mathbf{13}$

★ $d = \sqrt{9 + 9} = \mathbf{4.24}$

Examples 3 and 4 — Negative Coordinates

EXAMPLES · 5 MIN

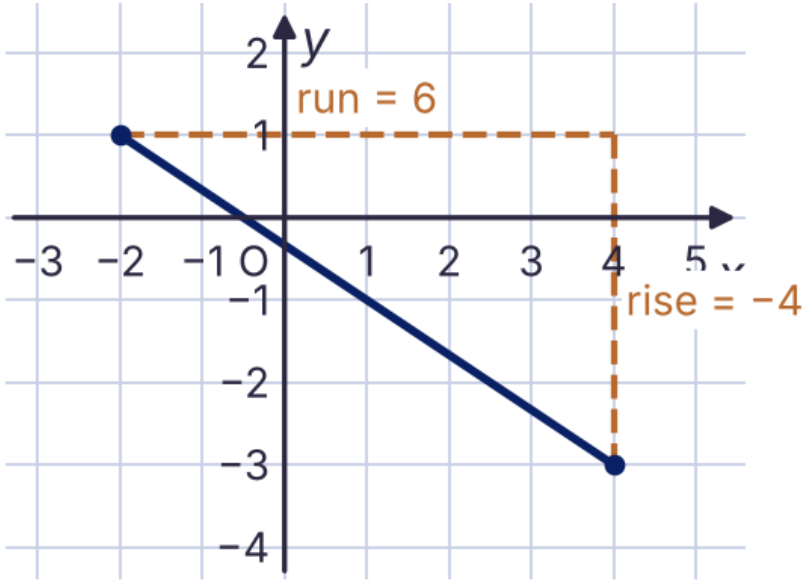
SILVER

Watch and copy into your book

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Example 3

Find the distance between (−2, 1) and (4, −3) to 3 s.f.



Example 4

Find the distance between (–5, –2) and (1, 6).

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Across = 6

Up = −4

$d = \sqrt{36 + 16}$

$d = \sqrt{52}$

= 7.21

Tip: squaring makes the change positive.

Example 4

Across = 6

Up = 8

$d = \sqrt{36 + 64}$

$d = \sqrt{100}$

= 10

Tip: subtract carefully: 6 − (−2) = 8.

Task 2

PRACTICE · 9 MIN

SILVER

Give answers to 3 s.f. where needed

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

1. (−1, 2) and (2, 6)
2. (−3, −1) and (5, 5)
3. (4, −2) and (−1, 3)
4. (−6, 0) and (0, −8)
5. (−2, −5) and (3, 4)

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $d = \sqrt{9 + 16} = \mathbf{5}$

2. $d = \sqrt{64 + 36} = \mathbf{10}$

3. $d = \sqrt{25 + 25} = \mathbf{7.07}$

4. $d = \sqrt{36 + 64} = \mathbf{10}$

5. $d = \sqrt{25 + 81} = \mathbf{10.3}$

Examples 5 and 6 — Distance Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Example 5

(3 marks)

A is (1, 2), B is (7, 2) and C is (4, 6).

Show that triangle ABC is isosceles.

Example 6

(3 marks)

The distance between (2, k) and (5, 1) is 5.

Find the possible values of k.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$AC = \sqrt{3^2 + 4^2}$$
$$= 5$$
$$BC = \sqrt{3^2 + 4^2}$$
$$= 5$$

AC = BC, so ABC is isosceles

Tip: find two sides and compare.

Example 6

$$3^2 + (k - 1)^2 = 25$$
$$(k - 1)^2 = 16$$
$$k - 1 = \pm 4$$

k = 5 or −3

Tip: square both sides of the distance.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

1. Find the length of AB, A(−3, 4), B(5, −2). (2 marks)

2. Find the perimeter of the triangle with vertices (0, 0), (6, 0) and (6, 8). (3 marks)

3. A circle has centre (2, 1) and passes through (5, 5). Find its radius. (2 marks)

4. Show that (1, 1), (4, 5) and (8, 2) make an isosceles triangle. (3 marks)

5. The distance between (1, 3) and (k, 7) is 5. Find the two values of k. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $d = \sqrt{64 + 36} = \mathbf{10}$

2. $6 + 8 + \sqrt{36 + 64} = 6 + 8 + 10 = \mathbf{24}$

3. $r = \sqrt{9 + 16} = \mathbf{5}$

4. $AB = \sqrt{9 + 16} = 5, \quad BC = \sqrt{16 + 9} = 5,$ **Isosceles**

5. $(k - 1)^2 + 16 = 25, \quad k - 1 = \pm 3, \quad k =$ **4 or −2**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the distance between (1, 1) and (7, 9)

SILVER Find the distance between (−2, 3) and (3, −9)

GOLD Find the distance between (0, 0) and (4, 6) to 3 s.f.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $d = \sqrt{36 + 64} = \mathbf{10}$

SILVER $d = \sqrt{25 + 144} = \mathbf{13}$

GOLD $d = \sqrt{16 + 36} = \mathbf{7.21}$

Exam Practice — Using Pythagoras

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

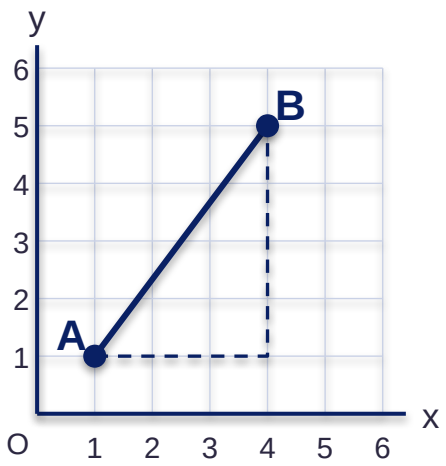
Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Question 1

(3 marks)

A is (1, 1) and B is (4, 5).

Work out the length of AB.



.....

Question 2

(3 marks)

P is (2, 3) and Q is (8, 11).

Work out the length of PQ.

.....

Exam Practice — Negative Coordinates

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Question 3

(3 marks)

Work out the distance between (−2, 1) and (4, −3).

Give your answer to 3 s.f.

.....

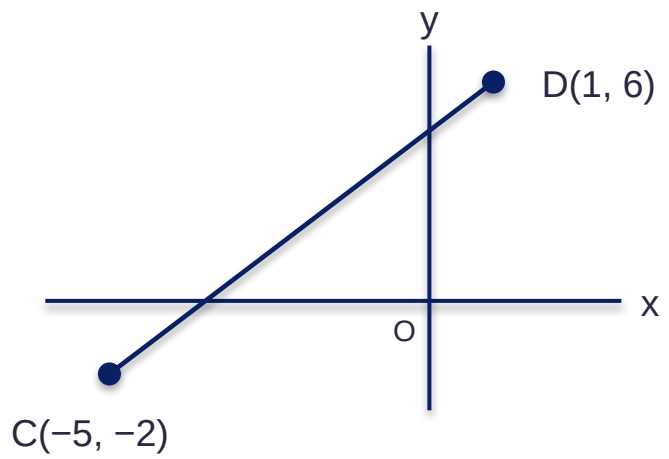
Question 4

(3 marks)

C is (−5, −2) and D is (1, 6).

Work out the length of CD.

Diagram NOT accurately drawn



.....

Exam Practice — Distance Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

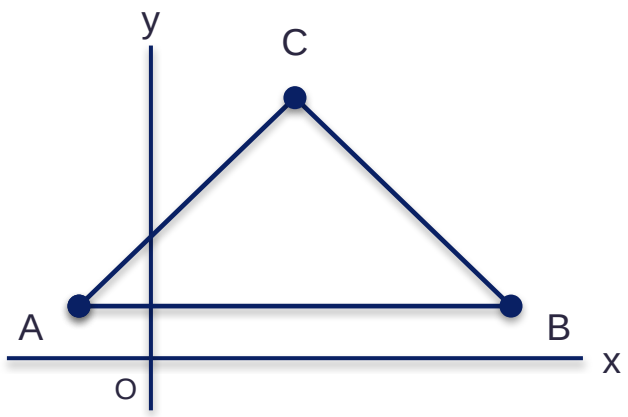
Question 5

(3 marks)

A is (−1, 1), B is (5, 1) and C is (2, 5).

Show that triangle ABC is isosceles.

Diagram NOT accurately drawn



Question 6

(4 marks)

The distance between (k, 3) and (1, −1) is 5.

Find the two possible values of k.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$\Delta x = 3,\ \Delta y = 4$	M1
$3^2 + 4^2 = 25$	M1
AB = 5	A1

Tip: draw the right-angled triangle.

Question 2

$\Delta x = 6,\ \Delta y = 8$	M1
$6^2 + 8^2 = 100$	M1
PQ = 10	A1

Tip: $d^2 = (\Delta x)^2 + (\Delta y)^2$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\Delta x = 6,\ \Delta y = -4$	M1
$36 + 16 = 52$	M1
$\sqrt{52} = 7.21$	A1

Tip: squaring removes the negative sign.

Question 4

$\Delta x = 6,\ \Delta y = 8$	M1
$36 + 64 = 100$	M1
CD = 10	A1

Tip: 1 − (−5) = 6.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$AC^2 = 3^2 + 4^2 = 25,\ AC = 5$ — M1

$BC^2 = 3^2 + 4^2 = 25,\ BC = 5$ — M1

AC = BC, so isosceles — A1

Tip: AB = 6, so only two sides are equal.

Question 6

$(k - 1)^2 + 4^2 = 25$ — M1

$(k - 1)^2 = 9$ — M1

$k - 1 = \pm 3$ — M1

$k = 4$ or $k = -2$ — A1

Tip: remember ± when square-rooting.